

# Requirements Definition

## Functional Requirements

### 1. User Authentication

- 1.1. Students can log in to the system.
- 1.2. Manager (Admin) can log into the system.
- 1.3. Users can change their passwords.

### 2. Student Management & Discipline

- 2.1. Manager can register a new student.
- 2.2. Manager can update student information.
- 2.3. Manager can delete student records.
- 2.4. Students can view his/her profile.
- 2.5. Manager can record and view student violations.

### 3. Hostel & Room Management

- 3.1. Manager can manage hostel information.
- 3.2. Manager can allocate rooms to students.
- 3.3. Manager can view available rooms and check maximum limits.
- 3.4. Students can view room details.

### 4. Complaint Management

- 4.1. Students can submit complaints mapped to their room.
- 4.2. Manager and Caretaker can view complaints.
- 4.3. Manager can review complaints and take necessary action.

### 5. Invoice & Payment Management

- 5.1. Manager can generate invoices.
- 5.2. Students can view invoice details.
- 5.3. Manager can update billing and payment status.

## **6. Attendance Management**

- 6.1. Manager can record student attendance history.
- 6.2. Manager can record staff attendance patterns.
- 6.3. Manager can view attendance history.

## **7. Staff Management**

- 7.1. Manager can add staff profiles.
- 7.2. Manager can update staff information.
- 7.3. Manager can manage staff salaries.
- 7.4. Manager can manage staff leaving records.
- 7.5. Manager can record staff violations.

## **8. Mess Management**

- 8.1. Students can apply for Mess Off for particular date ranges.
- 8.2. Manager can approve/view Mess Off requests.
- 8.3. Students can view the weekly mess menu.
- 8.4. Manager can update the weekly mess menu.

## **9. Visitor Management**

- 9.1. Manager can register student visitors.
- 9.2. Manager can view visitor tracking records.

## **10. Suggestions & Feedback**

- 10.1. Students can submit feedback records.
- 10.2. Manager can view suggestions.

## **11. Student In/Out Management**

- 11.1. Students can submit leave information.
- 11.2. Manager can monitor student entry and exit records.

## **12. Parcel Management**

- 12.1. Manager/Staff can track arriving parcels for students.
- 12.2. Students can track the collection status of their parcel.

## **Non-Functional Requirements**

### **1. Operational Requirements**

- 1.1. The system will operate as a web-based application accessible via modern web browsers.
- 1.2. The system should automatically back up all database records at the end of each day.
- 1.3. The system must generate invoice and attendance reports in a printable PDF format.
- 1.4. The parcel tracking and attendance monitoring modules should seamlessly interface with barcode/QR scanners or biometric hardware if integrated later.
- 1.5. The system should provide a responsive user interface for Students submitting complaints or checking menus.

### **2. Performance Requirements**

- 2.1. The system will retrieve and display available rooms, room allocation, and weekly mess menu details in 2 seconds or less.
- 2.2. The system will update billing and payment status within 3 seconds of the manager's confirmation.

### **3. Security Requirements**

- 3.1. Only authenticated users (Students, Managers) can access their respective dashboards after valid login.
- 3.2. Only a manager can generate invoices, delete student records, or manage staff salaries.
- 3.3. All user passwords must be securely encrypted and stored using industry-standard hashing algorithms.
- 3.4. Student profile data, particularly disciplinary and payment history, must be strictly confidential and visible only to the respective student and authorized management.

### **4. Cultural and Political Requirements**

- 4.1. No special cultural and political requirements are anticipated.